Docusign Envelope ID: D3EAC8FF-146A-4610-AFF4-26203B4BB02C

CONTINUED ON SHT 1 2000-F23-SLD-EL-40011

F23-SR-601
FIM III BOOSTER STATION
HV/LV SWITCHROOM

F23-SW-602
FIM III BOOSTER STATION
LV MOTOR CONTROL CENTRE

415V 3NE~ 50Hz 4000A 65kA for 1sec
EXTENDABLE

3F-13	3F-05	3F-09	4R-13	4R-17	1F-01	2F-01	1R-01	4F-10	3R-13	4R-21	
32A 160A	20A 160A	20A 160A	20A 160A	20A 160A	1250A 1250A	1250A 1250A	1250A 1250A	20A 160A	20A 160A	250A 250A	
2A 20A	2A 20A	2A 20A	2A 20A	2A 20A	2A 20A	2A 20A	2A 20A	2A 20A	2A 20A	2A 20A	
EOL	EOL	EOL	EOL	EOL				EOL	EOL		
I >	I >	I >	I >	I >	I >	I >	I >	I >	I >	I >	
NOTE 14	NOTE 14	NOTE 14	NOTE 14	NOTE 14	NOTE 15	NOTE 15	NOTE 15	NOTE 14	NOTE 14	NOTE 15	
M 3~	M 3~	M 3~	M 3~	M 3~	M 3~ RTD	M 3~ RTD	M 3~ RTD	M 3~	M 3~		
7.5kW	4kW	4kW	0.18kW	0.18kW	450kW	450kW	450kW	2.2kW	1.5kW	70A	
F23-HY-420	F23-FN-653	F23-FN-654	F23-FN-651	F23-FN-652	F75-PW-606	F75-PW-607	F75-PW-608	F75-PV-414	F75-PW-665	F75-SW-557	
FIM III HYDRAULIC POWER PACK	FIM III BOOSTER PUMP 1 MOTOR FAN	FIM III BOOSTER PUMP 2 MOTOR FAN	FIM III BOOSTER PUMP 1 GEARBOX FAN	FIM III BOOSTER PUMP 2 GEARBOX FAN	DECANT DAM 4 RETURN PUMP 1	DECANT DAM 4 RETURN PUMP 2	DECANT DAM 4 RETURN PUMP 3	DECANT DAM 4 VACUUM PUMP	FIM III BOOSTER STATION WASHDOWN PUMP	FIM III TSF SWITCHBOARD 2	
2000-F23-SCM-EL-40025	2000-F23-SCM-EL-40026	2000-F23-SCM-EL-40027	2000-F23-SCM-EL-40028	2000-F23-SCM-EL-40029	2000-F75-SCM-EL-40001	2000-F75-SCM-EL-40002	2000-F75-SCM-EL-40003	2000-F75-SCM-EL-40004	2000-F75-SCM-EL-40005	2000-F75-SCM-EL-40006	

NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC AND ON TRANSFORMER READING "M.E.N. LINK LOCATED IN F23-SW-602 INCOMER TIER" CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. UPSTREAM TRANSFORMER FEEDER SHALL BE TRIPPED ON ACTIVATION OF SWITCHROOM EMERGENCY PUSHBUTTON, DETECTION OF EARTH FAULT OR ARC FAULT DETECTION.
4. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
5. ALL DRIVE I/O VIA COMMS NETWORK TO DCS. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F23-BLK-IC-40001.
6. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
7. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
8. THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
9. TOLs TO BE MOUNTED IN PURPOSE-BUILT ENCLOSURE BENEATH VSD. TOLs MAY BE INCLUDED IN MCC MODULE WITH ENGINEERING APPROVAL.
10. ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
11. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
12. EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON N/C WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.
13. CURRENT LIMITING FUSE SIZE TO BE CONFIRMED BY SWITCHROOM VENDOR.
14. COMPATIBLE 'TD' TOROID TO MATCH NHP ELR.
15. COMPATIBLE 'EL500S' TOROID TO MATCH AMPCONTROL ELV.

LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 4, 5
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 4, 5
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 4, 5
- THERMAL OVERLOAD UNIT NOTE 8
- VARIABLE SPEED DRIVE NOTE 4, 5
- I > — EARTH LEAKAGE RELAY NOTE 4, 5
- FIELD-MOUNTED DECONTACTOR
- MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- MOTOR HEATERS (ONLY REQ. FOR MOTORS >90kW)
- 49 63 71 — TRANSFORMER PROTECTIVE DEVICES (REFER ANSI STANDARDS FOR DESCRIPTIONS)
- CIRCUIT BREAKER (NOTE 11)
- xxxA / xxxA — ST TRIP SET POINT / CB SIZE
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN

AS BUILT

REFERENCES	DWG No	REV	DATE	REVISION	DRN	CHK	TECH APP	APP
					MZ	JL		MB
		4	13.MAR.26	AS BUILT FOR FIM III TSF	[redacted]	EA	[redacted]	MBB
		3	07.JUL.25	RE-APPROVED FOR CONSTRUCTION	TG	[redacted]	KVN	
		2	17.MAR.25	RE-APPROVED FOR CONSTRUCTION	[redacted]	KVN	KVN	
F23-SW-602 LV MCC SINGLE LINE DIAGRAM SHT 1 OF 2	2000-F23-SLD-EL-40011	1	13.FEB.25	RE-APPROVED FOR CONSTRUCTION		[redacted]	KVN	
F23-SW-601 HV SWITCHBOARD SINGLE LINE DIAGRAM	2000-F23-SLD-EL-40001	0	17.JAN.25	APPROVED FOR CONSTRUCTION		KVN	[redacted]	

NORTHERN STAR RESOURCES LIMITED

DRA PACIFIC (PTY) LIMITED

[redacted]	[redacted]	11.NOV.2024
CHECKED	[redacted]	11.NOV.2024
DESIGNED	[redacted]	11.NOV.2024
TECH APP	[redacted]	26.NOV.2024
[redacted] APP	[redacted]	21.MAR.2025
CLIENT APP		

[redacted] KCGM
FIM TSF PROJECT

TITLE: FIM III BOOSTER STATION
F23-SW-602 LV MOTOR CONTROL CENTRE
SINGLE LINE DIAGRAM SHEET 2 OF 2

[redacted] | SCALE NTS | DRG No. 2000-F23-SLD-EL-40012 | REV 4